

PURPOSE

This Standard Operating Procedures (SOP) guide documents the repeatable processes, escalation paths, and operational runbooks for TechCorp’s IT Operations and Platform Engineering teams.

All on-call engineers and DevOps personnel are required to follow these procedures during normal operations and incident response.

SOP-001: PRODUCTION DEPLOYMENT PROCEDURE

Objective: Ensure safe, auditable, zero-downtime deployments to production environments.

Prerequisites:

- All automated tests pass in CI pipeline (GitHub Actions).
- Code review approved by at least 2 senior engineers.
- Change Request (CR) ticket created in ServiceNow with CR number.
- Deployment window: Monday-Thursday, 10:00 AM - 3:00 PM EST.
(No deployments on Fridays or weekends without CISO/CTO approval.)
- Deployment owner must be available for 2 hours post-deployment.

Steps:

1. Announce deployment in #deployments Slack channel with CR number.
2. Verify staging environment matches production configuration.
3. Run pre-deployment checklist (see Appendix A).
4. Execute deployment via ArgoCD (Kubernetes) or AWS CodeDeploy.
5. Monitor health dashboards for 30 minutes post-deployment.
6. Confirm all smoke tests pass.
7. Update ServiceNow CR ticket to "Deployed - Monitoring".
8. Post deployment summary in #deployments after 30-minute monitoring window.

Rollback Trigger: Any of the following triggers immediate rollback:

- Error rate > 1% of requests within 10 minutes.
- P95 latency exceeds 3x baseline.
- Any critical alert in PagerDuty during monitoring window.
- Customer-impacting bug reported via support channel.

Rollback Steps:

1. Execute rollback command in ArgoCD / CodeDeploy.
2. Notify #deployments and #incidents immediately.
3. Create P1/P2 incident in PagerDuty.
4. Initiate incident response per SOP-003.

SOP-002: DATABASE BACKUP & RECOVERY PROCEDURE

Objective: Ensure database integrity and recoverability.

Backup Schedule:

- Production RDS (PostgreSQL): Full backup daily at 02:00 UTC. Automated snapshots every 4 hours. Retention: 35 days.
- Analytics DWH (Snowflake): Daily export to S3 at 03:00 UTC. Retention: 90 days.
- MongoDB Atlas: Continuous backup with point-in-time recovery. Retention: 14 days.

Recovery Procedure (Database Restore):

1. Raise a P1 incident in PagerDuty; notify DBA on-call.